

Sandeep Kumar Patro CV

1. **Proposed Position** - Project V&V Leader (Expert)
2. **Name of Firm** - Alstom Transport India Limited
3. **Name** - SANDEEP KUMAR PATRO
4. **Date of Birth:** 14/06/1992 **Nationality:** INDIAN **Contact No:** 9439135529, 8904652962
Mail ID: sandeeppatro1405@gmail.com
5. **Education:**

Year	Degree	Institute
2016-18	MBA- Urban Management & Governance (Transportation)	Xavier University Bhubaneswar, Odisha
2009-13	B.Tech- Electronics & Telecommunication	Biju Patnaik University of Technology, Odisha

6. **Other Training:**
 - ☐ Certified **SCRUM MASTER** from Scrum Alliance - <https://bcert.me/svqernboh>
 - ☐ Certified **Alstom World Class Engineering** (WCE) in Signalling Validation.
7. **Expertise:** IXL, TMS, **Railway Signaling**, Intelligent Transportation Management system, Bid process management, Project Management
8. **Skills:** Validation & Verification, Project Engineering Management, Railway Signalling, Team Management, Signalling, Validation, Bid Management
9. **Countries of Work Experience:** INDIA, Denmark (FBANE), Philippines, Kiwi (New Zealand)
10. **Languages:**
 - English, Hindi, Odiya
11. **Employment Record:**

<i>Employer</i>	<i>Position Held</i>	<i>From</i>	<i>To</i>
Alstom Transport India Limited	Project V&V Lead (Expert)	August 2020	Till Now
PricewaterhouseCoopers Private Limited	Associate	January 2020	August 2020
KPMG Advisory Services Pvt. Ltd.	Consultant - Mobility	July 2019	January 2020
Ernst & Young LLP	Project Consultant	April 2018	April 2019
SIEMENS Rail Automation Pvt. Ltd.	Executive Engineer	August 2014	June 2016
BSBK Engineers Pvt. Ltd.	Engineer	September 2013	August 2014

Detailed Tasks Assigned

Company: ALSTOM INDIA TRANSPORT LIMITED

Action performed:

- Analyzing system requirement specification and writing test cases related to IXL and TMS.
- Creating SIMULATOR files based on requirement.
- Performing lab testing using VMs for IXL equipment.
- Managing system installation, testing & commissioning at site for TMS and Interlocking projects.
- Handling customer requirements and analyzing the need with team.
- Preparing verification and validation report, updating requirement verification table (RVT).
- Mapping and evaluating test cases are aligned with customer requirement.
- Preparing testing and commissioning checklist and strategy on how to complete T&C activity without any issue.
- Guiding development team on how the customer requirement can be fulfilled and what type of functionalities required.
- Updating Return on Experience (REX) on weekly basis and coaching team on how to troubleshoot using step by step method.
- Post test result, analyzing fault logs and preparing a summary on findings.
- Raising CRs with proper information and jointly analyzing logs with development team.
- Preparing release note, V&V report, delivery report for CCB meeting and explaining the updates about V&V and CRs to quality team.
- Coordinating with test engineers, V&V managers, Sub system Lead and project team about the requirement analysis.
- Travelling to on site for installation and testing activity and presenting functionality demo to customer.
- Coordinating with Business development team and design team to simplify the requirement of projects.
- Preparing requirement document and test report for site installation and testing activity and coordinating with quality team for Audit.
- Writing sub-system requirements derived from system requirements complete with tracing
- Writing interface specifications including documenting and describing the message formats
- Writing hardware design specifications including rough hardware drawings
- Contributing to the definition of the Program Verification and Validation strategy in the V&V Plan, and act proactively for its implementation;
- Defining verification methodology for selected deliverables, when requested in V&V strategy;
- When Safety is involved, share and justify the results with Safety Assurance Manager to feed the safety demonstration.
- Interfacing with experts in other department to align on selected V&V activity methodology;
- Represent Alstom in front of customers or partners, for instance to provide overview of test progress, issues, risks and opportunities;
- Specify, procure, develop, setup and configure assets such as testbenches and instrumentation;
- Provide data to maintain applicable KPIs;
- Manage configuration and change management cycle related with V&V activities (simulation configuration, anomaly tracking etc.);
- In Projects, follow-up with T&C for tests exported to site execution, Taking part to major developments by proving that the design is fit for its intended use, through implementation and execution of Verification, Integration and Test activities, fitting the targets in terms of Quality, Cost and Delivery.
- Performing all the in-house activities related with Verification, Integration and Testing performed by the V&V team
- Identifying if activities need to be exported to Projects or site execution, and follow up results. Interface with design teams to share and detail the results of V&V activities.

- Agree with Verification & Validation manager on strategy, scheduling and budget, and share progress, results and issues during execution.
- Contributing to the definition of the Project Verification and Validation strategy in the V&V Plan, and act proactively for its implementation
- Defining verification methodology for selected deliverables, when requested from V&V strategy
- Defining test plans and test descriptions for in-house activities. Liaise with PrCOM for exported tests
- Performing and reporting on allocated verification, integration and test activities
- Interface with experts in other department to align on selected V&V activity methodology
- Representing Alstom in front of customers or partners, for instance to provide overview of test progress, issues, risks and opportunities
- Performing various assessments to cross check the feasibility for each and every requirement raised by clients.
- Preparing functional testing reports and presenting to the clients
- Accountable for performance and the customer acceptance of the project solution.
- Manage the project technical risks and the corresponding mitigation plan
- Plans test activities for medium complexity projects, within the project schedules, and estimate the hours required to conduct all Test and Quality Assurance activities
- Creates test plans/protocols/procedures for each release. Assure all test documentation is complete prior to the start of the test cycle.

Company: PWC PVT. LTD.

Action performed:

- Took responsibility for the testing and commissioning of new and altered signalling systems for simple Works Testing of Signalling projects.
- Undertook testing duties within the scope of Certificate of Competency held.
- Undertook Work Package Management responsibilities on Projects.
- Identified resource requirements for your projects and competent staff are utilized, ensuring they are briefed regarding their work and responsibilities and that any necessary documentation is completed
- Cascaded management briefings as required.
- Ensured Audits and surveillances are undertaken in accordance with the Project Quality plan.
- Ensured adherence to the Railway safety standards, procedures, processes covering all engineering, safety, administrative and project requirements.
- Co-operate with quality, safety, and environmental objectives and contribute where appropriate to rail commitment for continual improvement.
- Managed testing and commissioning of signaling equipment to meet contract specifications, performance and safety parameters.
- Managed effective utilization and allocation of project-based testing resources both internal and subcontracted.
- Prepared and checked bids for accuracy and price for the testing portion of signaling projects
- Contributed to the continuous improvement of processes, procedures, innovations and initiatives.
- Encouraged and promoted the advancement of staff as a route to self-development

Company: KPMG ADVISORY SERVICES PVT. LTD.

Action performed:

- Inspected, evaluated, and performed simulated operational testing on assembled integrated electro-mechanical systems and equipment used in the railroad signal industry.
- Such equipment includes signal equipment enclosures, associated sub-components, wayside signals, switch machines, meter services, and power distribution cases used by railroads to institute automatic block signal systems, traffic control systems, train control and cab signal systems, interlocking systems, and rail and highway grade crossing protection systems.
- Performed simulated operational testing, circuit/logic troubleshooting, test documentation and customer acceptance testing of all types of signal systems.
- Performed logic simulation, rack testing and field commissioning of signal systems for multiple railroads

Company: ERNST & YOUNG LLP

Action performed:

- The delivery of high quality, innovative client solutions as part of a team
- Data processing, technical report writing, requirements gathering
- Project management support for small-to medium sized projects
- Assisting in the development of consultancy offerings, bids, and propositions
- Technical consultancy and assurance, and growing your area of technical expertise

Company: SIEMENS RAIL AUTOMATION PVT. LTD.

Action performed:

- Checking signaling principles designs, specifications, reports and working drawings to meet contract requirements within agreed timescales
- Perform troubleshooting, identification and resolution of system and subsystem issues and when required escalate issues to engineering and management
- Testing and documenting applications based on the projects functional and business requirements
- During project installation and testing time supported approval of client's schedules, invoices and pay requisitions, reviewed conformed signaling drawings & supported review of Request for Information (RFIs), submittals and field changes.
- Material Management:
Preparation of "Purchase Requisition" based on the contract "Purchase Order" Ensuring the receipt of materials at site as per the issued "Purchase order"
- "Material Reconciliation" of every station after commissioning & submission of Variation report to manager in-case of excess quantity installed.
- Planning and execution of Manpower required & work schedule at site.
- Verifying the Interface-Circuits and Co-ordination with design team for any alterations as per site condition/Customer requirement.
- Extending support to the bidding team in hardware/signaling gear estimation based on the tender stage signaling plans.
- Testing of Station Specific logic using GSIM (simulator).
- Testing (SAT) of logic with Actual system in connection with Signal's, Tracks & Block instruments.
- During commissioning time monitoring and documenting the status of the hardware and software configuration, to certify installation test, system test & supported project listing and close out.
- Coordinated with supply chain team to procure goods and deliver to clients on time.
- Performance of functional tests and commissioning as well as Reporting, documentation of work performed.
- Facilitated acceptance certification obtaining in a progressive manner, to ensure satisfactory testing and commissioning results are compiled and processed by supporting handover strategy.

Company: BSBK ENGINEERS PVT. LTD.

Action performed:

- Ensured safety compliance requirements are properly specified in product design requirements.
- Drive continuous improvement for systems engineering and related quality processes.
- Involved in hardware engineering of safety-related systems through knowledge of special requirements.
- Done Technical interface management by application of technical requirements management
- Done Validation in the form of analysis and tested with (sub-) system and systems and including checking the requirements for correctness focused on planning, conducting and documenting the validation
- Demonstrated, by analysis and testing, that the product meets the users' requirements, with respect to safety and quality as well as with emphasis on the suitability of its operation in accordance to its purpose in its intended environment.